Term 6 Checklist

Topic	Learning Outcomes	RAG	
		1	2
Y7 Key Points	Draw a labelled diagram of an animal cell including: membrane, nucleus and cytoplasm		
	Describe the role cell organelles e.g. membrane, nucleus, cytoplasm, mitochondria		
	Compare plant and animal cells using a diagram		
	Define communicable disease		
	State the types of organisms which cause disease		
	Suggest ways we can prevent the spread of communicable diseases		
	Give the word equation for aerobic respiration		
	Explain why heart rate must increase during intense exercise		
	Name the different forms of energy store e.g. kinetic, thermal		
	Draw energy transfer diagrams for a candle, a catapult and a TV		
	Identify useful and wasted energy in an energy transfer		
	Explain how fossil fuels are used to generate electricity in a power station		
	Compare the advantages/disadvantages of renewables and fossil fuels		
	Describe how carbon dioxide causes a greenhouse effect		
	State the approximate percentage of each gas in our atmosphere		
	Describe two processes that have caused the atmosphere to change over time		
	Describe what an element is in terms of types of atoms		
	Explain why chemical compounds are hard to separate		
	State the physical properties of the three states of matter		
	Describe how litmus paper can be used to test acids and bases		
	Describe how universal indicator can be used to test acids and bases		
	Compare how useful different pH indicators are		
	Explain why filtration could not be used to separate a mixture of salt and water		
	Describe how distillation can be used to separate a mixture of liquids		
	Describe the difference between contact and non contact forces		
	Draw force arrows on objects that are at rest or moving		
	Calculate the resultant force of an object when two forces are acting on it		
	Explain why your weight changes on different planets but your mass does not		
	Explain why friction occurs between two surfaces		
	State the equation that links speed, distance and time.		
	Calculate speed, distance or time using the equation s = d/t. Try this one: A car travels 90m in 45s. What is its speed?		
	Explain how pressure is caused inside a can of coke in terms of particles		
	Identify producers, consumers and decomposers in a feeding relationship		

	State the resources that animals and plants compete for		
	Describe the role of pollen and how plants reproduce		
	Give the word equation for photosynthesis		
	Explain how plants are adapted for photosynthesis		
	Describe the roles of different plant structures e.g. leaves, roots, stems, flowers		
	State the general features of the five major vertebrate groups		
	Describe what it means for an animal to be adapted to its environment		
	Describe how acid rain is formed from burning fossil fuels		
	Describe how charged particles interact		
	Compare electrical conductors and insulators		
	State the units of current and potential difference		
	Explain why current is the same at all points in a circuit		
	Draw complete circuit diagrams using a rule, pencil and straight lines		
	Draw a range of common circuit symbols e.g. wire, voltmeter, bulb		
	State what happens when two magnets are brought close together		
	Draw a magnetic field pattern around a bar magnet		
	Describe how to make an electromagnet stronger (use the 3cs)		
	Compare how a coal fired power station and wind turbine work		
Human Body	Draw labelled diagrams of plant and animal cells		
	Explain the importance of mitochondria in cells		
	Define diffusion and give examples of diffusion in the human body		
	Explain why starch is an important storage molecule		
	Describe how to perform an iodine test for starch and interpret the results		
	Name the major digestive organs and describe their functions		
	Describe the order food travels through the digestive system		
	Describe where chemical and mechanical digestion take place		
	Explain the importance of digesting the food we eat		
	Label the main parts of the lungs on a diagram		
	Explain the adaptations show by the alveoli		
	Explain the effect of smoking on lung function		
	Explain how breathing works in terms of pressure in the chest		
Heating and Cooling	Recall the three states of matter and the four state changes		
	State the boiling and freezing point of water		
	Explain what is meant by expansion and contraction		
	Describe what happens to the particles in a substance when it is heated		
	Describe how thermal energy transfers between hotter and colder objects		
	Give the standard units of temperature and energy		
	Compare temperature to thermal energy in terms of kinetic energy of particles		
	Describe how thermal energy transfer occurs through solids through conduction		
	Define the terms insulator and conductor with examples of each		

	Describe how thermal energy transfers through liquids and gases by convection		
	Describe how surfaces and temperature affect thermal radiation transfer		
	State three methods used to reduce heat transfer in a house		
	Explain insulation methods in terms of reducing conduction, convection or thermal radiation		
Periodic Table	Identify elements from the formula of a compound		
	Identify numbers of different atoms in compounds from a formula		
	Describe what a semi-metal is		
	Describe the properties of the transition metals		
	Use electronic structure to explain reactivity		
	Explain why you float in very salty water		
	Identify the group and period of an element on the periodic table		
	raw a labelled diagram of an atom including: nucleus, protons, neutrons, shells and electrons		
	State the charge of the three subatomic particles		
	State the names of groups 1, 2, 7 and 8 of the periodic table		
	Explain why displacement reactions occur		
	escribe the test and positive results for the gases: hydrogen, chlorine, carbon dioxide and oxygen		
	Describe how to collect gases using a submerged measuring cylinder		
Staying Alive	Compare asexual and sexual reproduction		
	Describe how fertilisation happens in animals		
	Describe the main changes that occur during puberty		
	Describe how puberty is controlled by hormones		
	Label the male reproductive organs		
	Label the female reproductive organs		
	Describe the function of the male reproductive system		
	Describe the function of the female reproductive system		
	Describe the steps involved in the menstrual cycle		
	Describe how oxygen and glucose reach a growing foetus		
	Explain the impact of smoking and vaping on foetal development		
	Describe the two causes of variation in organisms		
	Give examples of variation caused by environmental and genetic factor		
Restless Earth	Describe the transfer of heat through conduction, convection and thermal radiation		
	State the order of the planets in the Solar System		
	Explain why Pluto is no longer regarded as a planet		
	Describe the steps involved in the formation of the solar system from a gas cloud		
	Describe the Earth's orbit around the Sun		
	Describe the moon's orbit around the Earth		
	State the length of a day, lunar month and solar year		
	Explain how the tilt of the Earth is linked to seasons		
	Describe what is meant by an axis of rotation		

	Explain why it is Summer in the Northern hemisphere when it is Winter in the Southern hemisphere		
	Describe some impacts on the environment of a large asteroid impact		
	Explain why an asteroid impact can both cool or warm the Earth's atmosphere		
	Explain why Venus has higher average temperatures than Mercury		
	Describe the greenhouse effect using a diagram		
	Compare longitudinal and transverse waves		
	Draw a diagram showing the compression and rarefactions in a longitudinal wave		
	Define parallel and perpendicular		
	State the 7 parts of the electromagnetic spectrum		
	Describe the properties shared by all electromagnetic waves		
	Label wavelength and amplitude on a transverse wave		
	Describe the relationship between the frequency and wavelength of a wave		
	Draw diagrams to show how the pitch and volume of a sound affects the shape of the wave		
	Use a protractor to measure an angle of reflection		
	State the law of reflection		
	Label a diagram of the internal structure of the Earth		
	Explain why earthquakes happen due to tectonic plate movement		
	Describe how sedimentary, igneous and metamorphic rocks are formed		
Chemical Reactions	Give examples of chemical and physical changes		
	Describe the difference between chemical and physical changes in terms of particles		
	Describe changes of state in terms of forces between particles		
	Recall the formulae of common chemical compounds e.g. water, carbon dioxide, methane		
	Describe the difference between compounds and mixtures		
	Explain why the properties of elements change when they react together		
	Know what these parts of a chemical equation mean: reactants, arrow, products		
	Write chemical word equations when given the reactants and products		
	Predict the names of chemical compounds made from metals and non-metals		
	Predict the names of chemical compounds made from metals, non-metals and oxygen		
	Describe what is meant by thermal decomposition		
	Explain why the mass of a metal carbonate decreases when it is heated		
	State the general chemical word equation for the reaction of acids and metal oxides		
	Predict the products of reactions between metal oxides and acids		

	State the order of common metals in the reactivity series		
	Define the term displacement reaction		
	Predict whether a given metal and metal salt will react together using the reactivity series		
	Explain why gold and silver are used for jewellery in terms of reactivity		
	Explain why a solid precipitate is observed in the displacement of copper sulphate by iron (HT only)		
	Describe the difference between a solution and a chemical compound		
	Compare physical and chemical properties		
	Describe the properties of metals that make them useful in cooking pans and wires		
	Explain why metals are malleable and good conductors of electricity (HT only)		
	Write a chemical word equation for the rusting of iron		
	State factors which increase the rate of rusting		
	Explain why it is important to paint the outside of cars and boats		